

Lecture 25-Dynamic NAT & PAT (NAT Overload)

PART 1 — DYNAMIC NAT

What is Dynamic NAT?

Definition:

Dynamic NAT is a method where:

- Router automatically assigns **public IPs**
- From a **NAT pool**
- To internal (private) devices **only when needed**

Difference from Static NAT:

Feature	Static NAT	Dynamic NAT
Mapping	One-to-One, permanent	One-to-One, temporary
Assignment	Manual	Automatic
Exists when traffic flows	Always	Only when traffic exists

Key Concepts of Dynamic NAT

Important Points:

- Uses **ACL** to select which traffic to translate
 - Uses a **NAT pool** (range of public IPs)
 - Mapping is **one-to-one**
 - Entries **timeout automatically** if not used
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3 Role of ACL in Dynamic NAT 📄

ACL (Access Control List) decides **which internal IPs are translated**:

ACL Result NAT Action

PERMIT IP will be translated

DENY IP will **not** be translated (packet still forwarded)

⚠ Important: **DENY ≠ DROP**

4 NAT Pool Explained 🏊

A **NAT pool** is a **range of public IPs** used for translation.

Example:

NAT Pool: 100.0.0.1 – 100.0.0.2

- Router assigns **one public IP** per private host
 - Automatically from the pool
-

5 One-to-One Mapping (Dynamic NAT) 🔄

- Each internal IP gets **one public IP**
 - Same as Static NAT
 - Difference: **automatic assignment**
-

6 NAT Pool Exhaustion 🚨

Occurs when:

- All public IPs in the pool are **already used**
- A new host tries to access the Internet

Result:

- ❌ Router **drops the packet**
- ❌ Host **cannot access the Internet** until a public IP becomes free

7 Dynamic NAT Timeout ⌚

- Dynamic NAT entries **expire automatically** after inactivity
- Can be removed manually:

clear ip nat translation *

8 Daily Life Example (Dynamic NAT) 🏢🌐

- Office network: 50 PCs, 10 public IPs
- Only **10 users** can access Internet simultaneously
- Others must wait

Limitation: Dynamic NAT depends on **available public IPs**

◆ PART 2 — PAT (NAT OVERLOAD)

9 What is PAT? 1234 🌐

Definition:

- **Port Address Translation (PAT)** = NAT Overload
 - Translates **IP + PORT**
 - Allows **many devices** to share **one public IP**
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10 Why PAT is Important ⭐

Reason	Explanation
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IPv4 shortage	Only 1 public IP needed
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Scalability	Thousands of devices can connect
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Real-world use	Common in homes, offices, ISPs
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🏆 **PAT is the most widely used NAT type**

1 1 How PAT Works 🧠

- Router uses **same public IP** for multiple devices
- Differentiates using **port numbers**

Port numbers:

- 16-bit → ~65,000 ports available

Router keeps track of:

| Inside Local IP | Inside Global IP | Port Number |

1 2 Daily Life Example (PAT) 🏠 📊

Home router:

Device	Uses Public IP & Port
--------	-----------------------

Phone 📱	203.0.113.1:1024
---------	------------------

Laptop 💻	203.0.113.1:1025
----------	------------------

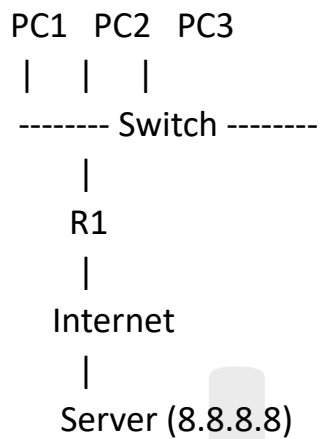
TV 📺	203.0.113.1:1026
------	------------------

Tablet 📱	203.0.113.1:1027
----------	------------------

- All share **one public IP**
 - Each device gets a **different port**
-

DYNAMIC NAT & PAT LAB (Step-by-Step)

Topology



IP Addressing

PCs:

Device IP

PC1 172.16.0.1

PC2 172.16.0.2

PC3 172.16.0.3

Gateway:

172.16.0.254

Router R1:

Interface IP

G0/1 172.16.0.254 (Inside)

G0/0 203.0.113.1 (Outside)

1 Configure Inside & Outside Interfaces 📁

R1> enable

R1# configure terminal

interface g0/1

ip nat inside

interface g0/0

ip nat outside

exit

2 Configure ACL (Select Traffic) 📄

access-list 1 permit 172.16.0.0 0.0.0.255

- Allows all internal PCs for NAT
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3 Create NAT Pool 🏊

ip nat pool Pool1 100.0.0.1 100.0.0.2 netmask 255.255.255.0

- Only **2 public IPs available**
 - Router assigns dynamically
-

4 Enable Dynamic NAT 🔁

ip nat inside source list 1 pool Pool1

- Uses ACL **1** and NAT Pool **Pool1**
-

5 Verify Dynamic NAT Table 🔍

show ip nat translations

Example output:

Inside Local: 172.16.0.1

Inside Global: 100.0.0.1

- IPs assigned **dynamically**
 - Mapping is **one-to-one**
-

6 Test NAT Pool Exhaustion ❌

- Ping from **PC1 & PC2** → **works**
 - Ping from **PC3** → **fails** (waiting)
 - Reason: **only 2 public IPs in pool**
-

7 Clear Dynamic NAT & Configure PAT ⚙️

Remove Dynamic NAT

no ip nat inside source list 1 pool Pool1

- **Inside/Outside interfaces** remain
- **ACL** remains

Configure PAT (Overload)

R1(config)# ip nat inside source list 1 pool Pool1 overload

OR

R1(config)# ip nat inside source list 1 interface g0/0 overload

- Uses **R1 G0/0 public IP**
 - Overloads using **ports**
-

8 Verify PAT Translations 🔍

show ip nat translations

Example output:

Inside Local: 172.16.0.1:1024

Inside Global: 203.0.113.1:1024

Protocol: ICMP

- **One public IP**, many ports
 - Many devices can access Internet simultaneously
-

Generate Traffic

From all PCs:

ping 8.8.8.8

- NAT table fills **only when traffic exists**
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Dynamic NAT vs PAT

Feature	Dynamic NAT	PAT (Overload)
Mapping	One-to-One	Many-to-One
IP Usage	Requires multiple public IPs	Only 1 public IP needed
Scalability	Low	Very High
Real World Use	Rare	Very Common
CCNA Importance	Medium	Very High
Port Usage	No ports	Uses port numbers
Traffic Limit	Limited by NAT pool	Can handle 65,000+ flows

Daily Life Difference Example:

- **Dynamic NAT:** Office with 10 public IPs → only 10 users online at a time
 - **PAT:** Home router → all devices use **1 public IP**, each gets a unique port
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Exam Summary

Topic	Key Points
Dynamic NAT	Automatic one-to-one translation from NAT pool
PAT / NAT Overload	Uses single public IP + port numbers
ACL	Selects which traffic to translate
NAT Pool	Range of public IPs
NAT Pool Exhaustion	Packet dropped if no IP available
Timeout	Dynamic NAT entries removed automatically
Commands	ip nat inside source list <ACL> pool <PoolName> ip nat inside source list <ACL> interface <intf> overload
Verification	show ip nat translations show ip nat statistics

1 CHAPTER SUMMARY

◆ PART 1 — Dynamic NAT

1 What is Dynamic NAT? 🤖🌐

Dynamic NAT is a method where:

- Router **automatically assigns public IPs**
- Uses a **NAT pool** of public IPs
- Assigns **only when internal devices initiate traffic**

Difference from Static NAT:

Feature	Static NAT	Dynamic NAT
Mapping	One-to-One, permanent	One-to-One, temporary
Assignment	Manual	Automatic
Exists	Always	Only when traffic exists

✅ Key Concepts:

- Uses **ACL** to decide which internal IPs get translated
- Uses a **NAT pool** (range of public IPs)
- **One-to-One mapping**
- **Entries expire automatically** if idle

2 Role of ACL in Dynamic NAT 📄

ACL decides which traffic is eligible for NAT.

ACL Result NAT Action

PERMIT IP will be translated

DENY IP is not translated (still forwarded)

⚠ Important: DENY does **not** drop the packet; it just excludes it from NAT translation.

3 NAT Pool Explained 🏠

- NAT pool = **range of public IPs** used for translation
 - Example: 100.0.0.1 – 100.0.0.2
 - Router assigns **one public IP per private host** automatically
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4 NAT Pool Exhaustion 🚨

Occurs when **all public IPs are used**.

Result:

- ❌ Router drops the packet
 - ❌ Host cannot access Internet until a public IP becomes free
-

5 Dynamic NAT Timeout ⌚

- Entries **expire automatically** after inactivity
- Can be cleared manually:

clear ip nat translation *

6 Daily Life Example (Dynamic NAT) 🏢🌐

- Office: 50 PCs, 10 public IPs
- Only 10 users can access Internet simultaneously
- Others must wait

Limitation: **Dependent on NAT pool size**

◆ PART 2 — PAT (Port Address Translation / NAT Overload)

1 What is PAT? 🧑🏫 🌐

- PAT = NAT Overload
 - Translates **IP + port number**
 - Allows **many internal devices to share a single public IP**
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2 Why PAT is Important ⭐

Reason	Explanation
--------	-------------

IPv4 shortage	Only 1 public IP needed
---------------	-------------------------

Scalability	Thousands of devices can connect
-------------	----------------------------------

Real-world use	Common in homes, offices, ISPs
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🏆 PAT is the **most widely used NAT type** in real-world networks.

3 How PAT Works 🧠

- Router uses **one public IP** for multiple devices
- Differentiates flows using **port numbers** (~65,000 available)
- Keeps a table:

| Inside Local IP | Inside Global IP | Port Number |

4 Daily Life Example (PAT) 🏠 📊

Device	Public IP + Port
--------	------------------

Phone 📱	203.0.113.1:1024
---------	------------------

Laptop 💻	203.0.113.1:1025
----------	------------------

TV 📺	203.0.113.1:1026
------	------------------

Device Public IP + Port

Tablet  203.0.113.1:1027

- **All share 1 public IP**
- **Each device gets a unique port**

Dynamic NAT vs PAT

Feature	Dynamic NAT	PAT (Overload)
Mapping	One-to-One	Many-to-One
IP Usage	Multiple public IPs	Only 1 public IP
Scalability	Low	Very High
Real World Use	Rare	Very Common
CCNA Importance	Medium	Very High
Port Usage	None	Uses ports
Traffic Limit	Limited by NAT pool	65,000+ flows
Daily Life Example	Office with 10 IPs → only 10 users online	Home router → all devices share 1 IP

CCNA Exam Key Points

Topic	Key Points
Dynamic NAT	Automatic one-to-one translation from NAT pool
PAT / NAT Overload	Uses single public IP + port numbers
ACL	Selects which traffic is translated
NAT Pool	Range of public IPs

Topic	Key Points
NAT Pool Exhaustion	Packet dropped if no IP available
Timeout	Dynamic NAT entries removed automatically
Commands	ip nat inside source list <ACL> pool <PoolName> ip nat inside source list <ACL> interface <intf> overload
Verification	show ip nat translations show ip nat statistics



Chapter Conclusion



Dynamic NAT:

- Assigns public IPs **automatically** from a pool
- Supports **one-to-one mapping**
- Limited by number of public IPs



PAT / NAT Overload:

- Assigns **one public IP** to many devices
- Uses **ports** to differentiate flows
- Highly **scalable**, suitable for home, office, ISP networks

Tip: PAT is the **most widely deployed NAT type in the real world.**

Q & A

1 What is Dynamic NAT?

Dynamic NAT automatically assigns a public IP from a **predefined pool** to internal devices **only when they initiate traffic**.

- Uses **ACL to select traffic**
 - Mapping is **temporary, one-to-one**
-

2 What is PAT / NAT Overload?

- Uses **single public IP** for multiple internal devices
 - Differentiates sessions using **port numbers**
 - Highly scalable, supports thousands of simultaneous connections
-

3 Difference: Static NAT vs Dynamic NAT vs PAT

Feature	Static NAT	Dynamic NAT	PAT
Mapping	One-to-One, permanent	One-to-One, temporary	Many-to-One
Assignment	Manual	Automatic	Automatic + ports
Public IPs	One per device	One per device	One IP for many devices
Scalability	Low	Medium	Very High
Use Case	Public servers	Limited dynamic devices	Home/office internet, ISPs

4 What is NAT Pool Exhaustion?

Occurs when **all public IPs in the pool are used**.

- Dynamic NAT: new devices cannot access internet until IP becomes free
 - PAT: avoids exhaustion by using **ports**
-

5 Why is ACL used in Dynamic NAT?

- **Selects which internal devices** get NAT translation
 - ACL PERMIT → translate
 - ACL DENY → do not translate (packet still forwarded)
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6 How do you configure Dynamic NAT?

```
access-list 1 permit 172.16.0.0 0.0.0.255
ip nat pool Pool1 100.0.0.1 100.0.0.2 netmask 255.255.255.0
ip nat inside source list 1 pool Pool1
```

7 How do you configure PAT?

```
ip nat inside source list 1 interface g0/0 overload
```

OR

```
ip nat inside source list 1 pool Pool1 overload
```

8 How to verify NAT / PAT?

```
show ip nat translations
```

```
show ip nat statistics
```

- Shows **inside local** → **inside global mapping**
 - Shows **port numbers** for PAT
-

9 What happens when NAT translation occurs?

- Router replaces **private IP with public IP** before sending traffic outside
 - For PAT: **Port number is added** to differentiate sessions
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10 Daily-Life Analogy

- **Dynamic NAT:** Office with limited public IPs → only some users online simultaneously
 - **PAT:** Home router → all devices share 1 IP using unique ports
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1 1 Practical Scenario Question

Scenario: 50 PCs, 10 public IPs, all trying to access internet

- **Dynamic NAT:** Only 10 users can access; 40 wait
 - **PAT:** All 50 users can access simultaneously using ports
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1 2 CCNA Exam Important Points

- Dynamic NAT requires **ACL + NAT pool**
 - PAT allows **many-to-one mapping using ports**
 - NAT pool exhaustion is a **common practical question**
 - Commands: `ip nat inside source list <ACL> pool <PoolName>` and `overload`
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